

CLASS 09 / OCT 20 / U05

LAYERED SITE ANALYSIS

P3d BEGINS — ONE SITE, MEASURED IN LAYERS

NOTHING DUE

P3d ISSUED TODAY / P3c CHECKPOINT TODAY / MID-SEMESTER DEADLINE NEXT CLASS — WEEK 10 — OCT 27

A SITE ANALYSIS IS A FIELD

Last class you built a field from distance. A site analysis reads the same way.

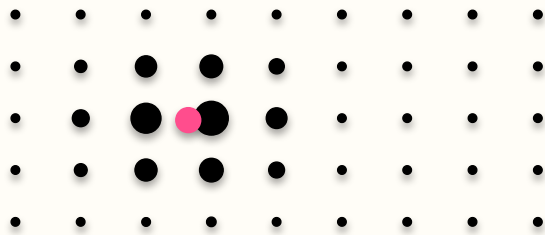
	ATTRACTOR FIELD — P3c	SITE LAYER — P3d
MEASURE	Distance to the attractor, at every point of the grid.	A value at every point of the site's point cloud.
REMAP	Distance range pushed into grey or a line property.	The value range pushed into grey or colour.
LEGEND	Near → far, with input and output ranges.	What the value means, with its range and units.
USE	The value drives a line in the drawing.	Layers combine into one reading of the site.

MEASURE → REMAP → LEGEND. THE SAME THREE STEPS, REPEATED FOR EVERY LAYER.

THREE KINDS OF LAYER

Each layer is a field: one value per point, its own range, its own legend.

DISTANCE TO POINTS OR CURVES

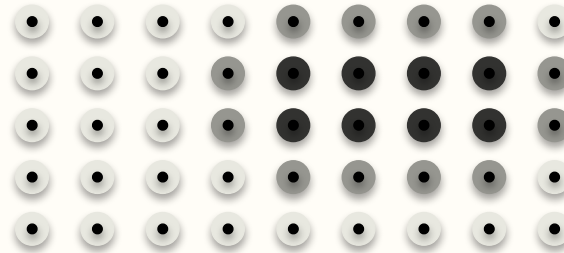


LEGEND · NEAR → FAR · m



Near → far from a feature you choose: a path, a tree line, an entrance.

HEIGHT BANDS

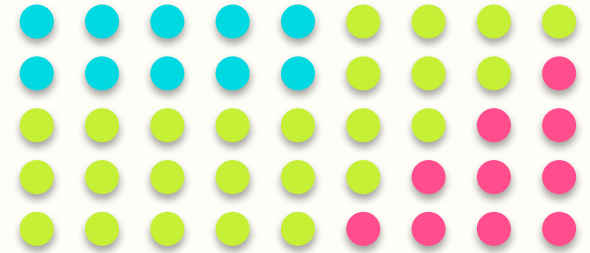


LEGEND · BAND EDGES · m



Every point sorted by its height into bands you set.

RECORDED ATTRIBUTE



LEGEND · CLASS OR VALUE



A value the scan already holds, such as colour or classification.

THREE LAYERS AT LEAST. EACH ONE ANSWERS A DIFFERENT QUESTION ABOUT THE SAME SITE.

Mixtli, the Blender add-on the instructor provides, does these operations. An equivalent tool is fine.

RECORD THE SITE FIRST

Before any layer, record the point cloud: where it came from and how it sits.

SOURCE

Where the point cloud came from. Name it on the sheet.

UNITS

What one unit is. Check it against one known dimension.

SCALE

The scale of each view, with a bar.

ORIENTATION

North, and how the cloud is placed relative to it.

AND ONE SENTENCE: WHY THIS SITE

A site you can document, and a reason you chose it. The reason tells you which layers to map.

P3d CHECKPOINT NEXT CLASS: BRING YOUR SITE'S POINT CLOUD AND AT LEAST TWO MAPPED LAYERS WITH LEGENDS.

COMBINE THE LAYERS INTO ONE READING

The tool computes each layer. How they combine, and what that shows, is your decision.

01 NORMALIZE

Remap each layer to 0–1 so the layers can be compared. Keep the original range and units on its legend.

02 WEIGHT

Decide how much each layer counts. Write the weights down.

03 COMBINE

Add, multiply, or keep the lowest or highest value. Say which, and why.

04 READ

Say what the combined view shows about the site that no single layer shows.

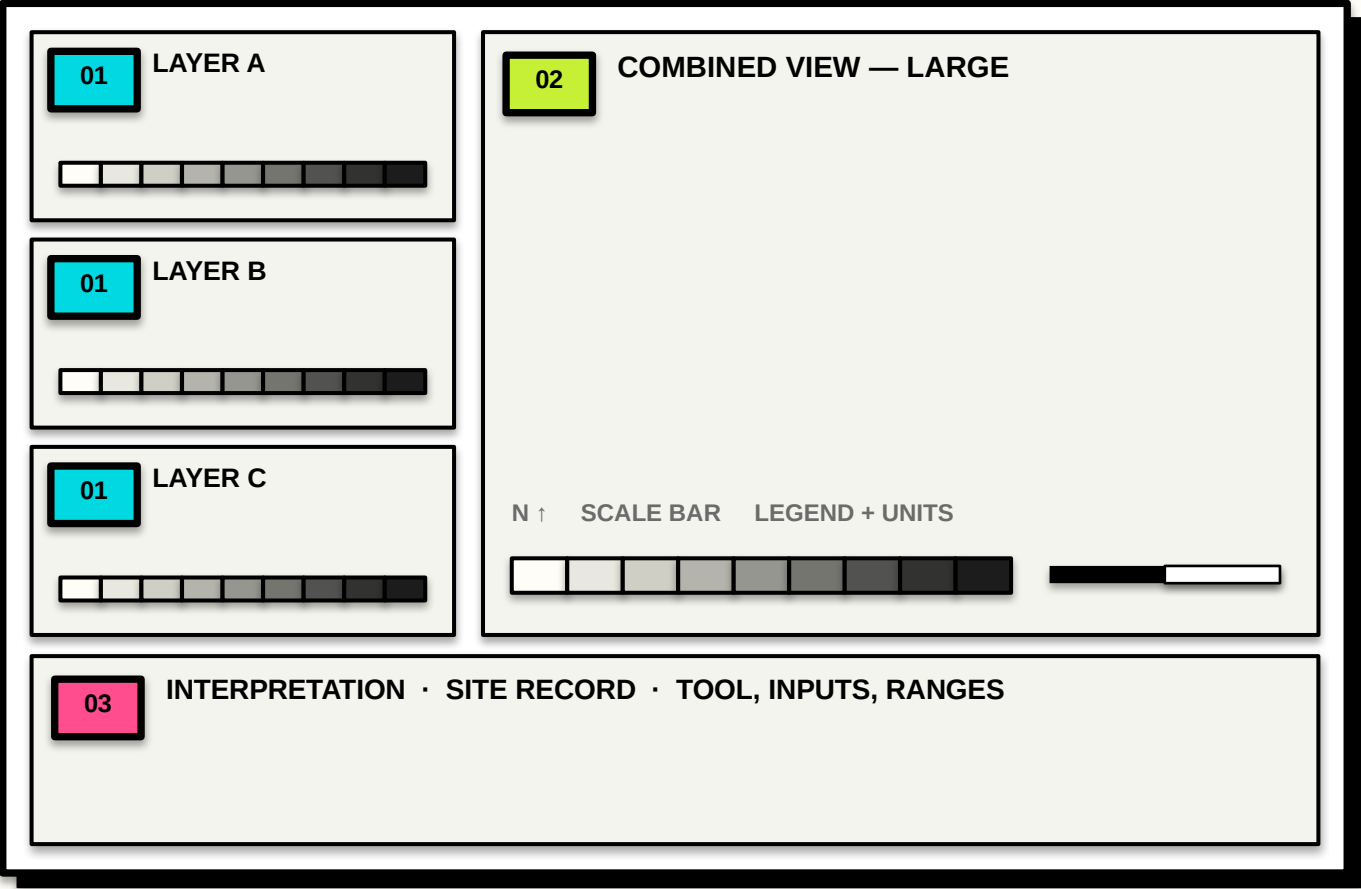
```
combined = w1 × distance + w2 × height + w3 × attribute    # w1 + w2 + w3 = 1
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SAME LOGIC AS P3c: A MEASURED VALUE, A STATED RANGE, A LEGEND. NOW THREE OF THEM.

A combined view with no layers beside it cannot be checked. A colourful image with no legend is not an analysis.

THE P3d SHEET — ONE 17 × 11

Small views of each layer beside one larger combined view. Every view has a legend.



EACH ZONE MUST SAY

- 01 Each layer: what it measures, its legend and units.
- 02 The combination: which layers, which weights, which operation.
- 03 What the combined view shows about the site — plus source, units, scale, north, and the tool you used.

SAME SCALE AND NORTH IN EVERY VIEW.

ISSUED TODAY

P3d — ATTRACTOR: LAYERED SITE ANALYSIS

DUE WEEK 11 — Nov 3 / 7%

Pick a site you can document and map its point cloud in at least three layers with attractor logic. Show each layer small, with its legend, beside one larger combined view, and say what the combination shows.

LAYERS

THREE AT LEAST

Distance to points or curves, height bands, a recorded attribute.

SHEET

ONE 17 × 11

Small views of each layer beside one larger combined view.

TOOL

MIXTLI OR EQUIVALENT

The Blender add-on the instructor provides.
Record inputs and ranges.

REQUIREMENTS — P3d

Visual quality 35% · computational understanding 25% · technical execution 20% · experimentation 10% · requirements and identity 10%.

01

Pick a site you can document and state why you chose it. Record the source of the point cloud, its units, scale and orientation.

02

Map at least three layers of the point-cloud data with attractor logic — for example distance to points or curves, height bands, and a recorded attribute such as colour or classification.

03

Make one 17 × 11 inch sheet with small views of each layer, each with its own legend and units, beside one larger combined visualization.

04

Explain what each layer measures, how you combined them, and what the combined view shows about the site.

05

Record the tool, inputs and ranges for every layer and submit the working file. The Mixtli add-on for Blender, introduced in class, does these operations; an equivalent tool is fine.

CHECKPOINT · WEEK 10 — Oct 27 — Bring your site's point cloud and at least two mapped layers with legends.

P3c CHECKPOINT — FIELD + FIRST LINES

Bring the grayscale field and a first line drawing of it. Read them side by side.

FIELD

Grayscale field with its legend:
near → far, input range written.

Wrong? Fix the remap before the lines.

LINES

First line drawing from the same field, with one mapping.

Wrong? Change one mapping value.

DENSITY

Darkest area still separate lines?
Lightest area still visible?

Wrong? Change the output range.

SVG

17 × 11 at 100%. Clean curves:
no fills, no duplicate lines.

Wrong? Fix the export.

PLOTTING IS OPTIONAL. IF YOU PLOT, TEST A SMALL CROP FIRST AND KEEP THE TEST FOR THE SHEET.

WEEK 10 DEADLINE CHECKLIST

Next class. Everything through Arrays must be in. Tick every line.

GSM	GRAPHIC STANDARDS MANUAL	Four pages. Each rule beside a tested application.	DUE NEXT CLASS
P2a	CSG PROCESS AND INSTRUCTIONS	The BIG-style process sheet, with instructions and pseudocode.	DUE W4 · IN?
P2b	CSG 3D PRINTED OBJECT	The print, its process sheet, and the source, mesh and slicer files.	DUE NEXT CLASS
P3a	DESCRIPTIVE GEOMETRY MODULE	One module sheet: studio render, one material, deliberate lighting.	DUE W6 · IN?
P3b	ARRAYS	Two sheets of arrays, with one conditional selection and its rule.	DUE W8 · IN?
P3c	ATTRACTOR: 2D FIELD DRAWING	One sheet: line drawing, grayscale field and legend. SVG and graph.	DUE NEXT CLASS

WEEK 10 — OCT 27 — MID-SEMESTER DEADLINE. GSM, P2b AND P3c ARE DUE. NOTHING THROUGH ARRAYS IS LEFT OUT.

NOW — CHECKPOINT, THEN LAYERS

Rest of the class: P3c review one table at a time, then start the site.

✓

P3c: field and first line drawing reviewed; one change written down

✓

P3d: site picked — source, units, scale and orientation recorded

✓

P3d: first layer mapped, with its legend and units

✓

P3d: the next two layers chosen, and what each will measure written down

✓

Deadline: every item on the checklist located and ready to submit

TWO LAYERS WITH LEGENDS BY NEXT CLASS.

BEFORE NEXT CLASS

DEADLINE

Next class, Week 10 — Oct 27, is the mid-semester deadline. GSM, P2b and P3c are due; everything through Arrays must be in.

NEXT

Week 10 — Oct 27 is a regular class: Volumetric Data — signed distance fields (SDF). P4a is issued.

P3d

Checkpoint next class. Bring your site's point cloud and at least two mapped layers with legends. P3d is due Week 11 — Nov 3.

P3c

Finish the sheet: line drawing, grayscale field and legend. Submit the SVG, the sheet PDF and the editable graph. Plotting is optional.